

Chapter 1

Triple Pendulum

In this chapter, we derive the Lagrangian of the triple pendulum, as well as explain how one can find all equations of motion of a triple pendulum.

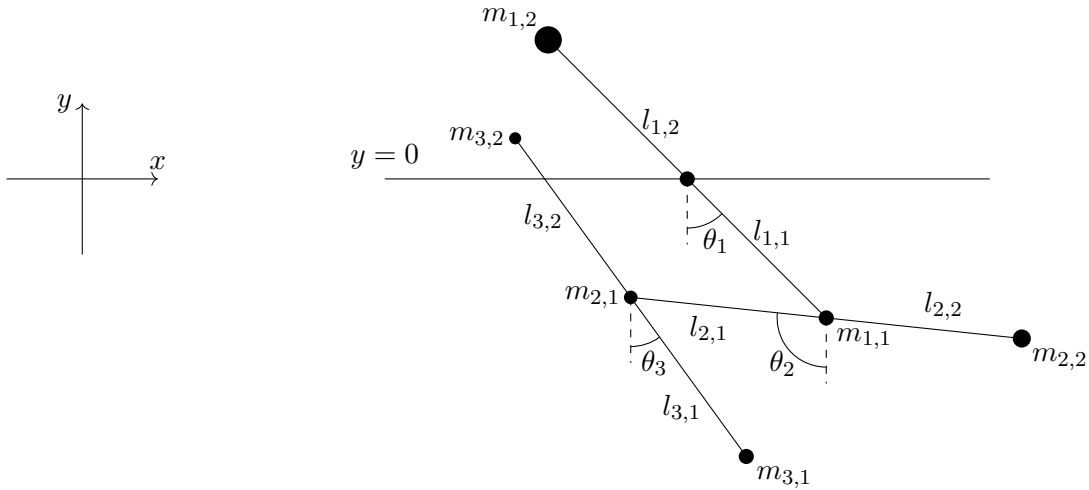


Figure 1.1: A triple pendulum with pivot points in the middle of the rod and masses on both ends

We first derive the Lagrangian of this system. It has 3 degrees of freedom. This can easily be seen, as we can express the whole system with the three angles θ_1, θ_2 and θ_3 . More formally, we have holonomic constraints on all the masses, since the distance from the mass to its pivot point always stays the same¹. We simplify the system a bit to make the equations a little bit shorter. We assume that $l_{n,m} = l_{n',m'}$, that is to say the rod is connected at the center to the pivot point and all lengths are the same. Furthermore, we assume the masses of the rod are all equal and homogeneously distributed. Recall the moment of inertia of a rotational object $E = \frac{1}{2} \cdot J \cdot \omega^2$, with $J := \int_K r^2 dm$. For a homogeneous thin rod width, we use the formula $J_s \approx \frac{1}{12} ml^2$. Now, we first calculate the value of the kinetic energy T . Notice that taking track of the minus signs is crucial in this system. We introduce a function s_m which is equal to -1 if the $m = 2$, and 1

¹for more details, see [1]

otherwise.

$$\begin{aligned}
 T &= \sum_{n \leq 3} \sum_{m \leq 2} \frac{1}{2} m_{n,m} (\dot{x}_{n,m}^2 + \dot{y}_{n,m}^2) + \sum_{n \leq 3} \frac{1}{2} \cdot J_n \cdot \omega_n^2 + \sum_{n \leq 3} \frac{1}{2} m_{\text{rod}} (\dot{x}_{\text{pivot},n}^2 + \dot{y}_{\text{pivot},n}^2) \\
 &= \sum_{n \leq 3} \sum_{m \leq 2} \frac{1}{2} m_{n,m} \left(\left(\frac{d}{dt} \left(\sum_{i < n} \sin(\theta_i) \cdot l + s_m \sin(\theta_n) \cdot l \right) \right)^2 + \left(-\frac{d}{dt} \left(\sum_{i < n} \cos(\theta_i) \cdot l + s_m \cos(\theta_n) \cdot l \right) \right)^2 \right) \\
 &\quad + \sum_{i \leq 2} \frac{1}{2} m_{\text{rod}} (\dot{x}_{i,1}^2 + \dot{y}_{i,1}^2) + \sum_{n \leq 3} \frac{1}{2} \cdot \frac{1}{12} m_{\text{rod}} \cdot l^2 \cdot \dot{\theta}_n^2
 \end{aligned}$$

We will now calculate the single terms and rearrange them:

$$\begin{aligned}
 T &= \frac{1}{2} (m_{1,1} + m_{1,2} + m_{2,1} + m_{2,2} + m_{3,1} + m_{3,2}) \dot{\theta}_1^2 l^2 + \frac{1}{2} (m_{2,1} + m_{2,2} + m_{3,1} + m_{3,2}) \dot{\theta}_2^2 l^2 \\
 &\quad + (m_{2,1} + m_{3,1} + m_{3,2} - m_{2,2}) l^2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + \frac{1}{2} (m_{3,1} + m_{3,2}) l^2 \dot{\theta}_3^2 \\
 &\quad + (m_{3,1} - m_{3,2}) l^2 \left(\dot{\theta}_1 \dot{\theta}_3 \cos(\theta_1 - \theta_3) + \dot{\theta}_2 \dot{\theta}_3 \cos(\theta_2 - \theta_3) \right) \\
 &\quad + \frac{1}{2} m_{\text{rod}} \left(2 \dot{\theta}_1^2 l^2 + \dot{\theta}_2^2 l^2 + 2 l^2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) + \frac{1}{24} \left(\dot{\theta}_1^2 + \dot{\theta}_2^2 + \dot{\theta}_3^2 \right) (2l)^2 m_{\text{rod}}
 \end{aligned}$$

Now, for the Lagrangian, we just subtract the potential term V and obtain:

$$\begin{aligned}
 L &= T - \left(gl \cos(\theta_1) (m_{1,2} - m_{1,1} - m_{2,1} - m_{2,2} - m_{3,1} - m_{3,2}) + gl \cos(\theta_2) (m_{2,2} - m_{2,1} - m_{3,1} - m_{3,2}) \right. \\
 &\quad \left. + gl \cos(\theta_3) (m_{3,2} - m_{3,1}) - gm_{\text{rod}} l (2 \cos(\theta_1) + \cos(\theta_2)) \right) .
 \end{aligned}$$

Recall the Euler-Lagrange equation² Plugging everything into the Euler-Lagrange equation and choosing $q_i = \theta_i$, we obtain the equation:

$$\begin{aligned}
 &\ddot{\theta}_1 \cdot l^2 \left(m_{1,1} + m_{1,2} + m_{2,1} + m_{2,2} + m_{3,1} + m_{3,2} + \frac{7}{6} m_{\text{rod}} \right) + \\
 &\ddot{\theta}_2 \cdot l^2 \cos(\theta_1 - \theta_2) (m_{2,1} + m_{3,1} + m_{3,2} - m_{2,2} + m_{\text{rod}}) + \\
 &\ddot{\theta}_3 \cdot l^2 \cos(\theta_1 - \theta_3) (m_{3,1} - m_{3,2}) = \\
 &(m_{2,1} + m_{3,1} + m_{3,2} - m_{2,2} + m_{\text{rod}}) l^2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2) + (m_{3,1} - m_{3,2}) l^2 \dot{\theta}_3 \sin(\theta_1 - \theta_3) (\dot{\theta}_1 - \dot{\theta}_3) \\
 &- gl \sin(\theta_1) (m_{1,1} + m_{2,1} + m_{2,2} + m_{3,1} + m_{3,2} - m_{1,2}) - 2gm_{\text{rod}} \sin(\theta_1) l - (m_{2,1} + m_{3,1} + m_{3,2} - m_{2,2}) \cdot \\
 &l^2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - (m_{3,1} - m_{3,2}) \dot{\theta}_1 \dot{\theta}_3 l^2 \sin(\theta_1 - \theta_3) - m_{\text{rod}} l^2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)
 \end{aligned}$$

Doing this for all 3 variables θ_1, θ_2 and θ_3 , we get a linear system, which we can solve for $\ddot{\theta}_1, \ddot{\theta}_2$ and $\ddot{\theta}_3$.

²for a rigorous derivation of the equation, see [1]

Bibliography

- [1] Simon J.A. Malham. An introduction to lagrangian and hamiltonian mechanics, 2016.